## Supplementary Material S5: Evaluation Protocol

This supplementary material documents the structured evaluation of the three NVL COPD sections modeled in this study. Each section is assessed against nine predefined criteria groups covering DMN decision structure, input data, output quality, FHIR terminological connectivity, BPMN process structure, roles, temporal aspects, exception handling, and inductive modeling findings. The nine criteria groups were derived from the seven benchmark dimensions established in Supplementary Material S1 (representation, decision logic, process logic, evidence linkage, interoperability, updateability, implementation proximity), operationalized specifically for BPMN 2.0 and DMN 1.3 as analytical intermediate models. The evaluation scale applied throughout is described in Table S5-1 and uses a four-value ordinal rating to distinguish directly modelable, partially modelable, non-modelable, and inductively identified findings.

**Table S5-1. Evaluation Scale**

| Value | Meaning |
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| ++ | Directly modelable; information is explicitly available. |
| + | Partially modelable; interpretation or assumption is required. |
| 0 | Not uniquely modelable; essential information is missing. |
| F | Additional finding that is not fully covered by the literature criteria. |

## Supplement S5a: Modelling and Interpretation Protocol for NVL COPD Section 3.2

Section 3.2 addresses tobacco cessation in patients with COPD. The evaluation focuses on the degree to which the guideline's recommendations regarding motivational assessment, consent to therapy, and selection of cessation interventions can be formally represented using BPMN and DMN. While the section provides a structured algorithm, several input variables lack operationalized measurement instruments, and the therapy loop contains no formal abort condition, requiring interpretive modeling decisions throughout.

**Table S5a-1. Evaluation protocol for NVL COPD Section 3.2.**

| Criteria Group & Focus | Assessment Question | Rating | Justification |
|---|---|---|---|
| DMN: Decision Structure | Are there clear if-then rules, decision points, negative rules? | + | It contains recognizable branches (consent yes/no, cessation successful yes/no). Six DMN rules could be derived. However, the central if-then logic for "low motivation" is not formalized. No score, no threshold value. The decision structure is described narratively and requires interpretation. |
| DMN: Input Data | Are variables measurable (laboratory/scores) and are threshold values defined? | 0 | None of the four DMN input variables is provided with a standardized measurement instrument.<br>• motivation_level is a clinical assessment without a score reference.<br>• consent_to_therapy and prior_cessation_attempt are documented states without an operationalized recording format. |

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| | | | • intensive_cbt_feasible depends on care structures that lie outside the guideline. |
| DMN: Output and Recommendation | Is the result unambiguous and provided with a strength of recommendation? | + | Combined therapy (CBT + pharmacotherapy) is supported with a strength of recommendation of ⇑⇑ (Recommendation 3-4). Alternatives such as brief counseling or online programs are marked with ⇑ (Recommendation 3-5).<br>• The output recommendation_grade serves informational purposes for documentation.<br>• The output intervention directly controls the assignment in the subsequent therapy sub-process. |
| FHIR: Terminology and Interface | Are medical standards (ATC, LOINC) used? Is it codable in FHIR? | 0 | The NVL mentions behavioral therapeutic and pharmacological measures without ATC codes or SNOMED-CT references. Motivational Interviewing is not coded in a standardized terminology system. A FHIR mapping is conceptually possible but requires a complete external terminology mapping without a guideline basis. |
| BPMN: Process Structure | Are start/end, sequence, and branches explicit? | + | A directed process flow with a recognizable start (COPD patient who smokes) and two end events (cessation successful / no further measures) is present. However, the sequence of steps is partly implicit.<br>• The start event is not a formal trigger condition.<br>• The return path in case of failed cessation is shown without an explicit abort condition or loop counter. |
| BPMN: Roles | Is it clearly defined who acts (physician, patient) and when roles change? | 0 | The NVL does not define explicit roles or swimlanes. Who documents the smoking habits is not specified.<br>• The division of roles (patient provides questionnaire data, physician completes documentation) was introduced as a modeling assumption.<br>• The timing of the role change during the consent query is also not formalized. |
| BPMN: Temporal Aspects | Are there deadlines, timer events, or repetition loops? | + | Abstinence of ≥ 6 months is named in the guideline text as an evidence endpoint, not as a procedural timer. An Intermediate Timer Event "After follow-up period" was introduced as an interpretation. No explicit repetition frequency or maximum number of therapy loops is defined. |
| BPMN: Exceptions and Abort | Are abort criteria, special cases, or escalation paths defined? | 0 | There is no formal abort criterion for the therapy loop.<br>• Neither a maximum number of attempts nor medical contraindications against pharmacotherapy are defined in the algorithm.<br>• According to the BPMN, the lack of consent leads to a return path, not to a process end. |
| Modeling Limits / Inductive Findings | Which gaps, ambiguities, or additional requirements only emerge during modeling? | F | F1 Lack of Operationalization of Motivation:<br>"Low motivation" is a matter of clinical judgment without a score or threshold value. DMN forces a binary classification (low/high), which excludes clinical intermediate stages.<br><br>F2 Care Structure as an Input Variable:<br>intensive_cbt_feasible depends on regional availability. A variable that lies outside the clinical process and is not formalized in any guideline. BPMN/DMN cannot map this dependency internally.<br><br>F3 Unlimited Therapy Loop: |

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| | | | The return path after failed cessation contains no loop counter, no maximum number, and no abort event. In BPMN, such an infinite loop is technically modelable but clinically not useful. A gap between guideline logic and procedural formalizability.<br><br>F4 Role Ambiguity in Documentation:<br>The guideline leaves open whether documentation is a medical or nursing task. The swimlane assignment in the BPMN is an unsubstantiated modeling assumption.<br><br>F5 Sub-Process without Internal Structure:<br>Combined therapy (CBT + pharmacotherapy) is modeled as a Collapsed Sub-Process because the NVL does not specify an internal structure for the measure. The internal process logic refers to an external guideline. A modeling limit due to a lack of guideline integration. |

## Supplement S5b: Modelling and Interpretation Protocol for NVL COPD Section 3.3

Section 3.3 addresses the use of e-cigarettes as a secondary cessation aid in patients with COPD who have previously failed or refused conventional tobacco cessation therapy. Unlike Sections 3.2 and 5.1, this section does not contain a graphical algorithm. The entire process flow was reconstructed from narrative text. The guideline formulates eligibility as a set of cumulative prerequisites rather than a classical recommendation with a graded strength, which required specific design decisions in both BPMN and DMN.

**Table S5b-1. Evaluation protocol for NVL COPD Section 3.3.**

| Criteria Group & Focus | Assessment Question | Rating | Justification |
|---|---|---|---|
| DMN: Decision Structure | Are there clear if-then rules, decision points, or negative rules? | + | The guideline contains a clear exclusion rule (primary cessation attempt → not suitable) and a cumulative positive rule ("only under the following prerequisites"). The AND-logic of the five eligibility conditions was modeled in DMN as a Hit-Policy-FIRST table with six rules.<br>• The if-then structure is formulated narratively and had to be interpretively converted into rule form.<br>• No formal decision tree exists in the guideline.<br>• The cascade of negation rules ensures that each mandatory criterion is checked in sequence. Only when all filters are passed does the final rule fire with a positive finding. |
| DMN: Input Data | Are variables measurable (labs/scores) and are thresholds defined? | 0 | All five DMN input variables are boolean and not measurable in a clinical sense.<br>• documented_failure_or_refusal, risk_counseling_completed, simultaneous_tobacco_cessation, and continuous_monitoring_planned are status variables without a standardized collection method.<br>• The guideline mentions neither a documentation format nor a measurement tool. |

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| | | | • Not a single threshold value or laboratory value is present in Section 3.3. |
| DMN: Output and Recommendation | Is the result unambiguous and provided with a strength of recommendation? | + | The DMN output "May consider under strict conditions" is deliberately formulated cautiously because the guideline does not assign a classic strength of recommendation (A/B) but rather describes a conditioned consideration.<br>• The exclusion rules ("Do not initiate") are clear.<br>• The positive eligibility output is not an operational command but a permission with reservations. A nuance that cannot be expressed natively in DMN. |
| FHIR: Terminology and Interface | Are medical standards (ATC, LOINC) used? Is it codable in FHIR? | 0 | E-cigarettes are not linked to medical standards in the guideline. Risk counseling, monitoring, and behavioral counseling are not standard-coded. A FHIR mapping would be conceptually possible but requires fully external terminology mapping without any guideline basis. This is the section with the lowest FHIR connectivity of all three modeled sections. |
| BPMN: Process Structure | Are start/end, sequence, and gateways explicit? | + | A clear exclusion path (primary cessation attempt → End Event) and a positive path (all prerequisites fulfilled → Initiate) can be derived from the text.<br>• The start is not explicitly formulated as a trigger; the triggering event (patient considers e-cigarette) had to be interpreted as a Message Start Event.<br>• The sequence of the four prerequisites in the sub-process is a modeling assumption, as the guideline provides only a list, not a process sequence. |
| BPMN: Roles | Is it clearly defined who acts (physician, patient) and when roles switch? | + | The guideline describes medical assessment duties (risk counseling, monitoring, eligibility check) implicitly, without explicit role assignment.<br>• The Patient pool was intentionally kept minimal, as Section 3.3 describes no patient actions other than triggering the need for counseling.<br>• The role switch from the Patient pool to the Healthcare Provider pool was modeled as a Message Flow. An interpretive but justifiable decision.<br>• A third lane (Follow-up/Monitoring) was added, even though the guideline does not name a separate role for monitoring. |
| BPMN: Temporal Aspects | Are there deadlines, timer events, or repetition loops? | 0 | "Continuous medical monitoring" is required, but neither the interval, duration, nor termination criteria are defined. The monitoring sub-process was modeled with a Timer Intermediate Event whose time value remains an unsubstantiated modeling assumption. There is not a single explicit time specification in Section 3.3. |
| BPMN: Exceptions and Abort | Are abort criteria, special cases, or escalation paths defined? | 0 | The monitoring sub-process ends at the gateway "Continue e-cigarette-supported abstinence?" The abort criteria for this gateway are not defined in the guideline.<br>• Neither adverse effects nor lack of abstinence are operationalized as formal abort conditions.<br>• Contraindications against e-cigarettes in COPD are not listed.<br>• The entire abort and escalation path is a model-side construction without a guideline basis. |
| Modeling Limits / Inductive Findings | Which gaps, ambiguities, or additional requirements emerge | F | F1 No Formal Algorithm Available:<br>Section 3.3 contains no graphical algorithm like Sections 5.1 or 3.2. The entire process flow was reconstructed from narrative. |

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| | only during modeling? | | F2 Continuous Boolean Simplification:<br>All five DMN inputs are boolean, even though clinical reality admits intermediate stages (e.g., partially documented failure, ongoing risk counseling). DMN forces a binary decision that does not match the gradual evidence level of the section.<br><br>F3 "Ideally" as a Non-Modelable Modality:<br>Behavioral counseling is qualified with "ideally". A modality between duty and option that cannot be expressed natively in BPMN. The Collapsed Sub-Process with annotation is a workaround, not a formal equivalent.<br><br>F4 Redundancy BPMN/DMN as a Design Decision:<br>This deliberate redundancy ensures independent comprehensibility of the DMN table but simultaneously shows that BPMN and DMN, without a shared execution model, do not guarantee that gateway conditions and decision rules will be evaluated identically at runtime<br><br>F5 Monitoring without Outcome Criteria:<br>The monitoring sub-process is formally modelable but empty in terms of content. Neither the abstinence criterion, side-effect threshold, nor termination decision are guideline-based. The gateway "Continue e-cigarette-supported abstinence?" is a structural necessity in BPMN but lacks guideline coverage. |

## Supplement S5c: Modelling and Interpretation Protocol for NVL COPD Section 5.1 (Exacerbations)

Section 5.1 covers pharmacological therapy escalation and de-escalation in patients with COPD who have experienced exacerbations. This section is the most formally specified of the three, featuring a graphical algorithm on p. 62 with explicit threshold values (eosinophil counts), hard abort criteria (pneumonia under ICS), and strong recommendation grades. It therefore achieved the highest overall modelability ratings and served as the structural anchor for the DMN cascade design.

**Table S5c-1. Evaluation protocol for NVL COPD Section 5.1 (Exacerbations).**

| Criteria Group & Focus | Assessment Question | Rating | Justification |
|---|---|---|---|
| DMN: Decision Structure | Are there clear if-then rules, decision points, negative rules? | ++ | The algorithm provided in section 5.1 contains explicit if-then structures: eosinophil thresholds, pneumonia as an abort condition, and asthma component as a protective rule. All three DMN tables (AdherenceCheck, IcsIndication, IcsReview) were derived in a rule-based and cascading manner. |

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| DMN: Input Data | Are variables measurable (laboratory/scores) and are threshold values defined? | + | Eosinophil values and pneumonia status are clearly measurable and are provided with hard threshold values.<br>• Adherence and inhalation technique are clinical assessments without a standardized measurement scale and had to be modeled as boolean variables. |
| DMN: Output and Recommendation | Is the result unambiguous and provided with a strength of recommendation? | ++ | The therapy recommendations (LAMA, LAMA+LABA, Triple Therapy, Roflumilast) are clearly named and supported with strong recommendation grades (⇑⇑). To correctly resolve the transitional grey area (100–299 cells/µl) described in the NVL, the logic was consolidated into a single catch-all rule (≥ 100) with the reason documented via the output variable. |
| FHIR: Terminology and Interface | Are medical standards (ATC, LOINC) used? Is it codable in FHIR? | 0 | The NVL names drug classes (LAMA, LABA, ICS) without ATC codes. Eosinophils are codable via LOINC but are not referenced in the guideline. Direct FHIR mapping requires an external terminology mapping without a guideline basis. |
| BPMN: Process Structure | Are start/end, sequence, and branches explicit? | ++ | The algorithm provided in section 5.1 represents a complete, directed decision tree with a defined start event ("Exacerbations present"), clear XOR gateways (Therapy-naive? / Prior treatment? / Roflumilast eligibility?), and unambiguous end events. |
| BPMN: Roles | Is it clearly defined who acts (physician, patient) and when roles change? | + | The NVL implicitly distinguishes between physician actions (therapy escalation, ICS review) and patient status. An explicit role assignment via lanes is absent from the guideline. The Physician/Patient lanes in the BPMN were added interpretively. |
| BPMN: Temporal Aspects | Are there deadlines, timer events, or repetition loops? | + | For the de-escalation path (ICS discontinuation), the guideline explicitly requires a follow-up visit after "a few weeks" in the advisory text of Section 5.1. This allows the guideline-compliant implementation of a BPMN Timer Event before re-evaluation. |
| BPMN: Exceptions and Abort | Are abort criteria, special cases, or escalation paths defined? | ++ | A pneumonia event during ongoing ICS therapy is operationalized in Recommendation 5-2 as a hard, protective abort criterion for steroid therapy and was modeled as a priority rule in DMN. |
| Modeling Limits / Inductive Findings | Which gaps, ambiguities, or additional requirements only emerge during modeling? | F | F1 Avoidance of Interval Redundancies:<br>To avoid logical gaps and ambiguous rule coverage with the FIRST Hit Policy, the intermediate and high eosinophil values were consolidated into a single rule (≥ 100 cells/µl). The transitional discretionary range (100–299) is documented via the output variable reason.<br><br>F2 Boolean Simplification:<br>Adherence and inhalation technique are gradual clinical assessments that had to be substantially simplified to boolean for DMN processing.<br>F3 Operationalization of the Discontinuation Time Window: |

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| | | | The imprecise time window ("a few weeks") for the follow-up visit after de-escalation necessitates a concrete temporal assumption in BPMN that is not exactly specified in the NVL.<br><br>F4 Roflumilast Criteria as Future DMN:<br>The conditions mentioned in the text for Roflumilast are represented in the current artifact as a BPMN gateway but should prospectively be extracted into a separate, fourth DMN table. |